

ℓ_1 -Embeddability of 6-Regular Klein-Bottle Graphs

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Abstract

We determine which finite simple 6-regular graphs on the Klein bottle have graph metrics that embed isometrically into ℓ_1 . Using Negami's classification and its comparison with Thomassen's notation, all such graphs are organized into three standard families, denoted $K_h(p, q)$, $K_{ce}(p, q)$ and $K_{co}(p, q)$. We prove that the unique isometrically ℓ_1 -embeddable graph among them is $K_{co}(5, 4)$, equivalently $H_{5,2,f}^*$ in Thomassen's notation. The non-exceptional cases are excluded by explicit five-point hypermetric certificates, while the positive case is realized by an explicit cut decomposition with eight cuts. As a quantitative refinement, the same certificates give explicit lower bounds for the least ℓ_1 -distortion $c_1(G)$ of every non-exceptional graph.

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1 Introduction

Let G be a finite connected graph, and let d_G denote its graph metric. We say that G is *isometrically ℓ_1 -embeddable* if there are an integer N and a map

$$\Phi : V(G) \longrightarrow \ell_1^N$$

such that

$$\|\Phi(x) - \Phi(y)\|_1 = d_G(x, y) \quad (x, y \in V(G)).$$

This paper studies isometric ℓ_1 -embeddability for finite simple 6-regular graphs on the Klein bottle. Negami's classification divides these graphs into three standard families

$$K_h(p, q), \quad K_{ce}(p, q), \quad K_{co}(p, q),$$

where K_h is the handle type, K_{ce} is the crosscap-even type, and K_{co} is the crosscap-odd type. Kasai, Matsumoto, Nakamoto, Nozawa, Seno and Takiguchi unified Negami's classification with Thomassen's classification; in particular,

$$K_h(p, q) = H_{p,q-1,a}^*, \quad K_{ce}(p, q) = H_{p,q-1,c}^*, \quad K_{co}(p, q) = H_{p,q-2,f}^*.$$

Moreover, the type $H_{k,m,b}^*$ in Thomassen's notation duplicates the type $H_{k,m,a}^*$. Thus Negami's three families cover all finite simple 6-regular Klein-bottle graphs [2, 4, 5].

The main contribution is a complete classification of the graph metrics in these families that lie in the finite ℓ_1 cone. The proof has two parts. First, we give explicit five-point hypermetric obstructions for all non-exceptional parameter values. Second, for the exceptional graph $K_{co}(5, 4)$, we give an explicit cut decomposition that realizes its graph metric as an ℓ_1 metric. We also record a quantitative consequence: every obstruction certificate gives a lower bound on the least ℓ_1 -distortion.

Theorem 1.1 (Main theorem). *Among finite simple 6-regular graphs on the Klein bottle, the only graph whose graph metric embeds isometrically into ℓ_1 is*

$$K_{co}(5, 4).$$

Equivalently, in Thomassen's notation, the unique positive example is $H_{5,2,f}^$.*

2 Standard models and the five-point inequality

We use the following triangular-lattice directions throughout:

$$(i, j) \sim (i, j + 1), \quad (i, j) \sim (i + 1, j), \quad (i, j) \sim (i + 1, j - 1),$$

with the second coordinate reduced modulo the corresponding value of p . If a source uses the opposite diagonal convention, replacing all second coordinates by their negatives converts it into the convention used here and does not affect the certificates below. Figure 1 illustrates the convention: all three labelled arrows follow actual edges of the local triangular lattice.

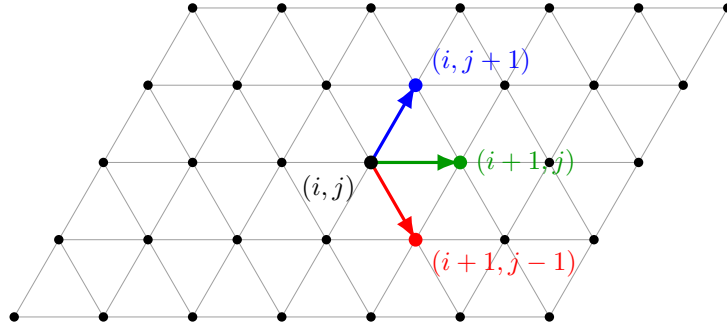


Figure 1: Local triangular-lattice coordinate rule. Edges from (i, j) go to $(i, j + 1)$, $(i + 1, j)$ and $(i + 1, j - 1)$.

Definition 2.1 (Standard coordinate models). *Let $\mathbb{Z}_p = \mathbb{Z}/p\mathbb{Z}$. The three families are represented as follows.*

- (i) $K_h(p, q)$, where $p \geq 3$ and $q \geq 3$: take $(\mathbb{Z} \times \mathbb{Z}_p)/\sim$, where

$$(i, j) \sim (i + q, -j),$$

and join each vertex to the equivalence classes of $(i, j + 1)$, $(i + 1, j)$ and $(i + 1, j - 1)$. Every vertex has a representative (i, j) with $0 \leq i < q$.

- (ii) $K_{ce}(p, q)$, where $p \geq 6$ is even and $q \geq 2$: start with the strip $0 \leq i \leq q$, $j \in \mathbb{Z}_p$, add the triangular-lattice edges above, and identify each boundary cycle by the half-turn

$$(0, j) \sim (0, j + p/2), \quad (q, j) \sim (q, j + p/2).$$

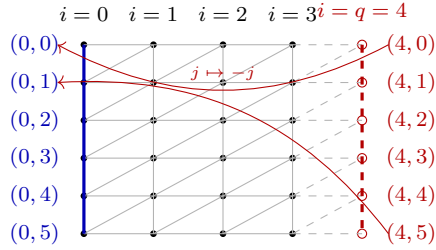
- (iii) $K_{co}(p, q)$, where $p = 2m + 1 \geq 5$ is odd and $q \geq 2$: the vertex set is

$$\{0, 1, \dots, q-1\} \times \mathbb{Z}_p.$$

In addition to the triangular-lattice edges inside the strip, add the crosscap edges

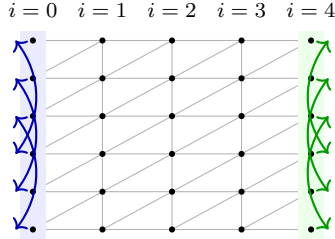
$$\begin{aligned} (0, j) &\sim (0, j + m), & (0, j) &\sim (0, j + m + 1), \\ (q-1, j) &\sim (q-1, j + m), & (q-1, j) &\sim (q-1, j + m + 1). \end{aligned}$$

The following schematic examples are placed immediately after the definition because they explain the boundary identifications used in the three standard models.



Example $p = 6$, $q = 4$: the solid columns $0 \leq i < q$ form the fundamental domain. The dashed copy $i = q$ is identified with $i = 0$ by $(q, j) \sim (0, -j)$.

Figure 2: Example of the handle family $K_h(p, q)$.



Example $p = 6$, $q = 4$: each boundary cycle is identified by the half-turn $j \mapsto j + 3$.

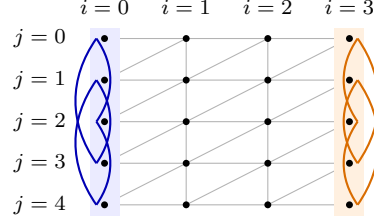
Figure 3: Example of the crosscap-even family $K_{ce}(p, q)$.

For five vertices a, b, c, s, t in a graph G , set

$$\begin{aligned} \Delta_G(a, b, c; s, t) &= d_G(a, b) + d_G(a, c) + d_G(b, c) + d_G(s, t) \\ &\quad - d_G(a, s) - d_G(a, t) - d_G(b, s) - d_G(b, t) - d_G(c, s) - d_G(c, t). \end{aligned}$$

Lemma 2.2 (Five-point hypermetric inequality). *If a finite metric space (V, d) embeds isometrically into ℓ_1 , then for every $a, b, c, s, t \in V$,*

$$\Delta(a, b, c; s, t) \leq 0.$$



Example $p = 5$, $q = 4$, $m = 2$: crosscap chord edges are added in the two boundary columns.

Figure 4: Example of the crosscap-odd family $K_{co}(p, q)$.

Proof. We use only the finite, discrete form of the cut decomposition. Let

$$\Phi = (f_1, \dots, f_N) : V \rightarrow \ell_1^N$$

be an isometric embedding, so that

$$d(x, y) = \sum_{r=1}^N |f_r(x) - f_r(y)|.$$

Fix one coordinate function $f = f_r$. Since V is finite, the set of values of f can be written as

$$f(V) = \{\lambda_0 < \lambda_1 < \dots < \lambda_m\}.$$

For $1 \leq k \leq m$, define the cut

$$C_k = \{v \in V : f(v) \geq \lambda_k\}.$$

If $f(x) = \lambda_i$ and $f(y) = \lambda_j$ with $i \leq j$, then exactly the cuts $C_{i+1}, C_{i+2}, \dots, C_j$ separate x and y . Hence

$$|f(x) - f(y)| = \sum_{k=1}^m (\lambda_k - \lambda_{k-1}) \delta_{C_k}(x, y),$$

where $\delta_C(x, y) = 1$ if C separates x and y , and 0 otherwise. Applying this to every coordinate gives a finite nonnegative linear combination of cut metrics:

$$d(x, y) = \sum_{r=1}^N \sum_{k=1}^{m_r} (\lambda_{r,k} - \lambda_{r,k-1}) \delta_{C_{r,k}}(x, y).$$

Therefore it is enough to check the inequality for a single cut metric.

Let $x_1 = a, x_2 = b, x_3 = c, x_4 = s, x_5 = t$, and put

$$\beta_1 = \beta_2 = \beta_3 = 1, \quad \beta_4 = \beta_5 = -1.$$

Thus $\sum_{i=1}^5 \beta_i = 1$, and the expression Δ is

$$\sum_{1 \leq i < j \leq 5} \beta_i \beta_j d(x_i, x_j).$$

For a cut $C \subseteq V$, set

$$\alpha = \sum_{i: x_i \in C} \beta_i.$$

The contribution of δ_C to Δ is the sum over separated unordered pairs, namely

$$\sum_{\substack{1 \leq i < j \leq 5 \\ x_i \in C, x_j \notin C}} \beta_i \beta_j + \sum_{\substack{1 \leq i < j \leq 5 \\ x_i \notin C, x_j \in C}} \beta_i \beta_j = \sum_{\substack{i: x_i \in C \\ j: x_j \notin C}} \beta_i \beta_j = \alpha \left(\sum_{j: x_j \notin C} \beta_j \right) = \alpha(1 - \alpha).$$

Because α is an integer, $\alpha(1 - \alpha) \leq 0$. Since d is a finite nonnegative linear combination of such cut metrics, the same inequality holds for d . $\square$

Consequently, a single five-point set with $\Delta_G(a, b, c; s, t) > 0$ proves that the graph metric of G is not isometrically ℓ_1 -embeddable. The tables below give such certificates with $\Delta_G = 1$. For independent checking, Appendix A lists the ten distances used in every certificate, and Appendix B gives a short verification script.

3 Excluding the non-exceptional graphs

Proposition 3.1. *No graph $K_h(p, q)$ is isometrically ℓ_1 -embeddable.*

Proof. The parameter ranges in Table 1 are disjoint and exhaustive for $p \geq 3$ and $q \geq 3$. Each row gives five vertices $a, b, c \mid s, t$. The ten graph distances used in each row are recorded in Table 4; substituting those distances into the definition of Δ_G gives

$$\Delta_{K_h(p, q)}(a, b, c; s, t) = 1$$

for every row. Lemma 2.2 then excludes an isometric ℓ_1 embedding. $\square$

Parameter range	Certificate $a, b, c \mid s, t$	Δ
$q = 2r, r \geq 2$	$(0, 0), (0, 1), (r, 0) \mid (1, 0), (r + 1, 0)$	1
$q = 3, p = 3$	$(0, 0), (1, 1), (2, 2) \mid (0, 1), (0, 2)$	1
$q = 3, p = 4$	$(0, 0), (1, 1), (2, 2) \mid (0, 1), (2, 1)$	1
$q = 3, p = 5$	$(0, 0), (1, 1), (2, 4) \mid (0, 1), (1, 0)$	1
$q = 3, p \geq 6, c = \lceil p/2 \rceil$	$(0, 0), (0, 2), (1, c) \mid (1, 2), (1, p - 1)$	1
$q = 5, p = 3$	$(0, 0), (1, 1), (2, 2) \mid (0, 1), (1, 2)$	1
$q = 5, p = 4$	$(0, 0), (0, 2), (2, 1) \mid (0, 1), (0, 3)$	1
$q = 5, p = 5$	$(0, 0), (1, 1), (3, 3) \mid (0, 2), (1, 0)$	1
$q = 5, p = 6$	$(0, 0), (1, 1), (3, 3) \mid (0, 2), (1, 5)$	1
$q = 5, p = 7$	$(0, 0), (0, 4), (2, 1) \mid (0, 3), (4, 1)$	1
$q = 5, p \geq 8$	$(0, 0), (0, \lfloor p/2 \rfloor), (2, 0) \mid (0, 2), (2, p - 1)$	1
$q = 7, p = 3$	$(0, 0), (1, 1), (2, 2) \mid (0, 1), (1, 2)$	1
$q = 7, p = 4$	$(0, 0), (0, 2), (2, 1) \mid (0, 1), (0, 3)$	1
$q = 7, p \geq 5$	$(0, 0), (0, 2), (3, 1) \mid (1, 1), (5, 0)$	1
$q = 2r + 1 \geq 9, p = 3$	$(0, 0), (1, 1), (2, 2) \mid (0, 1), (1, 2)$	1
$q = 2r + 1 \geq 9, p \geq 4$	$(0, 0), (0, 2), (r, 0) \mid (2, 0), (r + 3, 0)$	1

Table 1: Five-point obstruction certificates for $K_h(p, q)$.

Proposition 3.2. *No graph $K_{ce}(p, q)$ is isometrically ℓ_1 -embeddable.*

Proof. Take

$$a = (0, 0), \quad b = (0, 1), \quad c = (q, 0), \quad s = (1, 0), \quad t = (1, p/2).$$

In $K_{ce}(p, q)$,

$$d(a, b) = 1, \quad d(a, c) = q, \quad d(b, c) = q, \quad d(s, t) = 2,$$

and

$$\begin{aligned} & d(a, s) + d(a, t) + d(b, s) + d(b, t) + d(c, s) + d(c, t) \\ &= 1 + 1 + 1 + 1 + (q - 1) + (q - 1) = 2q + 2. \end{aligned}$$

The lower bounds for $d(a, c)$ and $d(b, c)$ follow because every path from the left boundary to the right boundary must change the first coordinate by q , and each edge changes it by at most one; the matching upper bounds are the evident straight paths in the strip. The equality $d(s, t) = 2$ is realized by moving from $(1, 0)$ to the left boundary class represented by both $(0, 0)$ and $(0, p/2)$ and then back to $(1, p/2)$; the two vertices s and t are not adjacent. The four distances $d(a, s), d(a, t), d(b, s), d(b, t)$ are equal to one by the boundary identification and triangular-lattice edges, while $d(c, s) = d(c, t) = q - 1$ follows from the same first-coordinate argument and straight paths. Thus

$$\Delta_{K_{ce}(p, q)}(a, b, c; s, t) = (1 + q + q + 2) - (2q + 2) = 1.$$

Lemma 2.2 proves the proposition. $\square$

Proposition 3.3. *Except for $K_{co}(5, 4)$, no graph $K_{co}(p, q)$ is isometrically ℓ_1 -embeddable.*

Proof. Write $p = 2m + 1$. The parameter ranges in Table 2 cover all odd $p \geq 5$ and all $q \geq 2$, except for $(p, q) = (5, 4)$. Each row gives five vertices $a, b, c \mid s, t$. The ten graph distances used in each row are recorded in Table 6; substituting them into the definition of Δ_G gives

$$\Delta_{K_{co}(p, q)}(a, b, c; s, t) = 1.$$

Lemma 2.2 completes the proof. $\square$

Parameter range	Certificate $a, b, c \mid s, t$	Δ
$p = 5, q = 2, 3$	$(0, 0), (0, 1), (q - 1, 2) \mid (0, 2), (1, 0)$	1
$p = 5, q \geq 5$	$(0, 0), (1, 1), (4, 2) \mid (0, 1), (3, 2)$	1
$p = 2m + 1 \geq 7, 2 \leq q \leq p - 2,$ $j = \max\{2, m - q + 2\}$	$(0, 0), (0, 1), (q - 1, j) \mid (0, m + 1), (1, 0)$	1
$p = 7, q \geq 6$	$(0, 0), (0, 2), (0, 4) \mid (0, 1), (0, 3)$	1
$p = 2m + 1 \geq 9, q \geq p - 1, \alpha = m - 3,$ $\beta = \max\{0, m - 5\}$	$(0, 0), (0, 2), (\alpha, 4) \mid (0, 1), (\beta, 4)$	1

Table 2: Five-point obstruction certificates for $K_{co}(p, q)$.

4 The positive example $K_{co}(5, 4)$

It remains to show that $K_{co}(5, 4)$ is indeed isometrically ℓ_1 -embeddable. Its vertex set is

$$V = \{0, 1, 2, 3\} \times \mathbb{Z}_5.$$

Define three cuts

$$A_0 = \{0\} \times \mathbb{Z}_5, \quad A_3 = \{3\} \times \mathbb{Z}_5, \quad M = \{0, 1\} \times \mathbb{Z}_5,$$

and, for each $r \in \mathbb{Z}_5$,

$$S_r = \{(i, r+t) : i = 0, 1, 2, 3, 0 \leq t \leq 3-i\},$$

where the second coordinate is reduced modulo 5. Thus S_r has column sizes 4, 3, 2, 1. Let

$$\mathcal{C} = \{A_0, A_3, M, S_0, S_1, S_2, S_3, S_4\}.$$

Proposition 4.1. *The graph $K_{co}(5, 4)$ embeds isometrically into ℓ_1^8 .*

Proof. Define

$$\Phi(v) = \frac{1}{2}(\mathbf{1}_C(v))_{C \in \mathcal{C}} \in \mathbb{R}^8.$$

Then

$$\|\Phi(x) - \Phi(y)\|_1 = \frac{1}{2} \sum_{C \in \mathcal{C}} \delta_C(x, y),$$

where $\delta_C(x, y) = 1$ if and only if C separates x and y . Put

$$N(x, y) = \sum_{C \in \mathcal{C}} \delta_C(x, y).$$

It is enough to show that

$$N(x, y) = 2d_{K_{co}(5,4)}(x, y) \quad (x, y \in V).$$

Consider the automorphisms

$$\rho_r(i, j) = (i, j + r), \quad \sigma(i, j) = (i, -j - i), \quad \tau(i, j) = (3 - i, -j),$$

where $r \in \mathbb{Z}_5$. Checking the edge types in Definition 2.1 shows that these maps preserve the edge set of $K_{co}(5, 4)$. They also send the cut family $\mathcal{C}$ to cuts in $\mathcal{C}$ or to their complements: ρ_1 fixes A_0, A_3, M and sends S_r to S_{r+1} ; σ fixes A_0, A_3, M and permutes the S_r ; τ exchanges A_0 and A_3 , sends M to its complement, and sends every S_r to the complement of some $S_{r'}$. Hence both $d(x, y)$ and $N(x, y)$ are constant on the unordered-pair orbits generated by these automorphisms.

The action has 16 orbits on the $\binom{20}{2} = 190$ unordered vertex pairs. Table 3 gives representatives of these orbits, and each row satisfies $N(x, y) = 2d(x, y)$. Therefore the displayed equality holds for every pair x, y , and the proposition follows. $\square$

5 Proof of the main theorem

Proof of Theorem 1.1. By Negami's classification, together with the comparison of Negami's and Thomassen's classifications by Kasai et al., every finite simple 6-regular graph on the Klein bottle belongs to one of the three families $K_h(p, q)$, $K_{ce}(p, q)$ and $K_{co}(p, q)$. Propositions 3.1–3.3 show that all graphs in these families, except possibly $K_{co}(5, 4)$, violate the five-point inequality of Lemma 2.2. Hence none of them embeds isometrically into ℓ_1 . Proposition 4.1 gives an explicit isometric ℓ_1 embedding of $K_{co}(5, 4)$. Finally, since $K_{co}(p, q) = H_{p,q-2,f}^*$, this graph is $H_{5,2,f}^*$ in Thomassen's notation. $\square$

Representative pair x, y	$d(x, y)$	$N(x, y)$
$(0, 0), (0, 1)$	1	2
$(0, 0), (0, 2)$	1	2
$(0, 0), (1, 0)$	1	2
$(0, 0), (1, 1)$	2	4
$(0, 0), (1, 2)$	2	4
$(0, 0), (2, 0)$	2	4
$(0, 0), (2, 1)$	3	6
$(0, 0), (2, 4)$	2	4
$(0, 0), (3, 0)$	3	6
$(0, 0), (3, 1)$	4	8
$(0, 0), (3, 3)$	3	6
$(1, 0), (1, 1)$	1	2
$(1, 0), (1, 2)$	2	4
$(1, 0), (2, 0)$	1	2
$(1, 0), (2, 1)$	2	4
$(1, 0), (2, 2)$	3	6

Table 3: Cut separation count and graph distance in $K_{co}(5, 4)$.

6 Quantitative remarks on ℓ_1 -distortion

The preceding theorem determines exactly when the least ℓ_1 -distortion equals one. We now record a quantitative consequence of the same certificates.

For a finite connected graph G , define

$$c_1(G) = \inf_{\Phi} \left(\sup_{x \neq y} \frac{\|\Phi(x) - \Phi(y)\|_1}{d_G(x, y)} \right) \left(\sup_{x \neq y} \frac{d_G(x, y)}{\|\Phi(x) - \Phi(y)\|_1} \right),$$

where the infimum is taken over all injective maps $\Phi : V(G) \rightarrow \ell_1^N$ and all finite dimensions N . For a finite set X , let $\text{CUT}(X)$ denote the cut cone on X , namely the set of all functions

$$\rho = \sum_{S \subseteq X} \lambda_S \delta_S \quad (\lambda_S \geq 0),$$

where $\delta_S(x, y) = 1$ if exactly one of x, y lies in S , and $\delta_S(x, y) = 0$ otherwise. Equivalently, after a uniform rescaling, one may write

$$c_1(G) = \inf \{ D \geq 1 : \exists \rho \in \text{CUT}(V(G)), \\ d_G(x, y) \leq \rho(x, y) \leq D d_G(x, y) \quad \forall x, y \in V(G) \}.$$

This is the standard finite-dimensional cut-cone description of ℓ_1 metrics; see, for example, Deza and Laurent [1]. The study of graph metrics through low-distortion embeddings is a central theme in the geometry of graphs [3].

Proposition 6.1 (Five-point distortion bound). *Let G be a finite graph and let $a, b, c, s, t \in V(G)$. Put*

$$P = d_G(a, b) + d_G(a, c) + d_G(b, c) + d_G(s, t)$$

and

$$Q = d_G(a, s) + d_G(a, t) + d_G(b, s) + d_G(b, t) + d_G(c, s) + d_G(c, t).$$

If $P > Q$, then

$$c_1(G) \geq \frac{P}{Q}.$$

In particular, if the certificate has $\Delta_G(a, b, c; s, t) = 1$, then

$$c_1(G) \geq 1 + \frac{1}{Q}.$$

Proof. Assume that an ℓ_1 metric ρ satisfies

$$d_G(x, y) \leq \rho(x, y) \leq Dd_G(x, y) \quad (x, y \in V(G)).$$

Since ρ is an ℓ_1 metric, Lemma 2.2 applied to ρ gives $P_\rho \leq Q_\rho$, where P_ρ and Q_ρ are obtained from the definitions of P and Q by replacing d_G with ρ . On the other hand, $P_\rho \geq P$ and $Q_\rho \leq DQ$. Therefore $P \leq DQ$, so $D \geq P/Q$. Taking the infimum over D gives the result. If $\Delta = P - Q = 1$, then $P/Q = (Q + 1)/Q = 1 + 1/Q$. $\square$

Corollary 6.2. *Every non-exceptional graph covered by Propositions 3.1–3.3 satisfies $c_1(G) > 1$. More precisely, for each certificate in Tables 1 and 2, and for the certificate in Proposition 3.2, the lower bound is*

$$c_1(G) \geq 1 + \frac{1}{Q},$$

where Q is the sum of the six cross-distances displayed in the corresponding row of Appendix A.

For example, the certificate for $K_{ce}(p, q)$ has

$$P = 1 + q + q + 2 = 2q + 3, \quad Q = 2q + 2,$$

and hence

$$c_1(K_{ce}(p, q)) \geq \frac{2q + 3}{2q + 2} = 1 + \frac{1}{2q + 2}.$$

The same bound also applies to the rows of Table 6 whose distance vector is

$$(1, q, q, 2; 1, 1, 1, 1, q - 1, q - 1).$$

These lower bounds are not asserted to be sharp. Determining the exact values, or even sharp asymptotics, of

$$c_1(K_h(p, q)), \quad c_1(K_{ce}(p, q)), \quad c_1(K_{co}(p, q))$$

for the non-exceptional parameters is a natural continuation of the present classification. For fixed p, q , the exact value can be formulated as a linear program over cut metrics:

$$\begin{aligned} & \text{minimize} && D \\ & \text{subject to} && d_G(x, y) \leq \sum_{S \subseteq V(G)} \lambda_S \delta_S(x, y) \leq Dd_G(x, y), \quad x, y \in V(G), \\ & && \lambda_S \geq 0. \end{aligned}$$

The number of cut variables is exponential in $|V(G)|$, but the symmetries of the three Klein-bottle families suggest substantial reductions. In particular, the eight-cut representation of $K_{co}(5, 4)$ in Proposition 4.1 is precisely a feasible solution with $D = 1$.

A Distance vectors for the five-point certificates

This appendix gives checkable details for the distance computations. If a certificate is written as $a, b, c \mid s, t$, the distance vector is ordered as

$$D = (d_{ab}, d_{ac}, d_{bc}, d_{st}; d_{as}, d_{at}, d_{bs}, d_{bt}, d_{cs}, d_{ct}).$$

Substitution into

$$\Delta = d_{ab} + d_{ac} + d_{bc} + d_{st} - d_{as} - d_{at} - d_{bs} - d_{bt} - d_{cs} - d_{ct}$$

gives 1 in every row.

Table 4: Distance vectors for the $K_h(p, q)$ certificates.

Parameter range	Distance vector D
$q = 2r, r \geq 2$	$(1, r, r, r; 1, r - 1, 1, r - 1, r - 1, 1)$
$q = 3, p = 3$	$(2, 2, 2, 1; 1, 1, 1, 1, 1, 1)$
$q = 3, p = 4$	$(2, 2, 2, 2; 1, 1, 1, 1, 2, 1)$
$q = 3, p = 5$	$(2, 2, 2, 1; 1, 1, 1, 1, 1, 1)$
$q = 3, p \geq 6, f = \lfloor p/2 \rfloor, c = \lceil p/2 \rceil$	$(2, f, c - 1, 3; 2, 1, 1, 2, c - 2, f - 1)$
$q = 5, p = 3$	$(2, 2, 2, 2; 1, 1, 1, 1, 2, 1)$
$q = 5, p = 4$	$(2, 3, 2, 2; 1, 1, 1, 1, 2, 2)$
$q = 5, p = 5$	$(2, 3, 3, 2; 2, 1, 1, 1, 2, 2)$
$q = 5, p = 6$	$(2, 3, 4, 3; 2, 1, 1, 2, 3, 2)$
$q = 5, p = 7$	$(3, 3, 3, 4; 3, 1, 1, 3, 2, 2)$
$q = 5, p \geq 8, f = \lfloor p/2 \rfloor$	$(f, 2, f, 3; 2, 2, f - 2, f - 1, 2, 1)$
$q = 7, p = 3$	$(2, 2, 2, 2; 1, 1, 1, 1, 2, 1)$
$q = 7, p = 4$	$(2, 3, 2, 2; 1, 1, 1, 1, 2, 2)$
$q = 7, p \geq 5$	$(2, 4, 3, 3; 2, 2, 1, 2, 2, 2)$
$q = 2r + 1 \geq 9, p = 3$	$(2, 2, 2, 2; 1, 1, 1, 1, 2, 1)$
$q = 2r + 1 \geq 9, p \geq 4$	$(2, r, r, r; 2, r - 2, 2, r - 2, r - 2, 3)$

Parameter range	Distance vector D
$p \geq 6$ even, $q \geq 2$	$(1, q, q, 2; 1, 1, 1, 1, q - 1, q - 1)$

Table 5: Distance vector for the $K_{ce}(p, q)$ certificate.

Table 6: Distance vectors for the $K_{co}(p, q)$ certificates.

Parameter range	Distance vector D
$p = 5, q = 2, 3$	$(1, q, q, 2; 1, 1, 1, 1, q - 1, q - 1)$
$p = 5, q \geq 5$	$(2, 4, 4, 4; 1, 3, 1, 3, 4, 1)$
$p = 2m + 1 \geq 7, 2 \leq q \leq p - 2$	$(1, q, q, 2; 1, 1, 1, 1, q - 1, q - 1)$
$p = 7, q \geq 6$	$(2, 1, 2, 2; 1, 1, 1, 1, 1, 1)$

Parameter range	Distance vector D
$p = 9, q \geq 8$	$(2, 2, 3, 2; 1, 1, 1, 2, 2, 1)$
$p = 2m + 1 \geq 11, q \geq p - 1$	$(2, m - 2, m - 1, m - 2; 1, m - 3, 1, m - 3, m - 2, 2)$

B Verification script

The following script constructs the finite graphs directly from Definition 2.1 and computes graph distances by breadth-first search. It can be used to verify Tables 1–3. The script uses only the Python standard library.

```

1 from collections import deque
2
3
4 def bfs(adj, source):
5     dist = {source: 0}
6     q = deque([source])
7     while q:
8         u = q.popleft()
9         for v in adj[u]:
10             if v not in dist:
11                 dist[v] = dist[u] + 1
12                 q.append(v)
13     return dist
14
15
16 def delta(adj, a, b, c, s, t):
17     D = {x: bfs(adj, x) for x in {a, b, c, s, t}}
18     return (D[a][b] + D[a][c] + D[b][c] + D[s][t]
19             - D[a][s] - D[a][t] - D[b][s] - D[b][t]
20             - D[c][s] - D[c][t])
21
22
23 def add_edge(adj, u, v):
24     if u != v:
25         adj[u].add(v)
26         adj[v].add(u)
27
28
29 def Kh_graph(p, q):
30     def norm(i, j):
31         n, r = divmod(i, q)
32         if n % 2:
33             j = -j
34         return (r, j % p)
35
36     V = [(i, j) for i in range(q) for j in range(p)]
37     adj = {v: set() for v in V}
38     for i, j in V:
39         u = norm(i, j)
40         for ni, nj in [(i, j + 1), (i + 1, j), (i + 1, j - 1)]:
41             add_edge(adj, u, norm(ni, nj))
42     return adj
43
44
45 def Kce_graph(p, q):
46     assert p % 2 == 0 and p >= 6 and q >= 2
47     h = p // 2
48
49     def norm(i, j):
50         j %= p
51         if i in (0, q):
52             j %= h

```

```

53         return (i, j)
54
55     V = []
56     for i in range(q + 1):
57         rng = range(h) if i in (0, q) else range(p)
58         V += [(i, j) for j in rng]
59     adj = {v: set() for v in V}
60     for i in range(q + 1):
61         for j in range(p):
62             u = norm(i, j)
63             add_edge(adj, u, norm(i, j + 1))
64             if i < q:
65                 add_edge(adj, u, norm(i + 1, j))
66                 add_edge(adj, u, norm(i + 1, j - 1))
67     return adj
68
69
70 def Kco_graph(p, q):
71     assert p % 2 == 1 and p >= 5 and q >= 2
72     m = (p - 1) // 2
73     V = [(i, j) for i in range(q) for j in range(p)]
74     adj = {v: set() for v in V}
75     for i, j in V:
76         add_edge(adj, (i, j), (i, (j + 1) % p))
77         if i < q - 1:
78             add_edge(adj, (i, j), (i + 1, j))
79             add_edge(adj, (i, j), (i + 1, (j - 1) % p))
80     for j in range(p):
81         for i in (0, q - 1):
82             add_edge(adj, (i, j), (i, (j + m) % p))
83             add_edge(adj, (i, j), (i, (j + m + 1) % p))
84     return adj
85
86
87 def Kh_witness(p, q):
88     import math
89     if q % 2 == 0:
90         r = q // 2
91         return (0, 0), (0, 1), (r, 0), (1, 0), (r + 1, 0)
92     if q == 3:
93         if p == 3:
94             return (0, 0), (1, 1), (2, 2), (0, 1), (0, 2)
95         if p == 4:
96             return (0, 0), (1, 1), (2, 2), (0, 1), (2, 1)
97         if p == 5:
98             return (0, 0), (1, 1), (2, 4), (0, 1), (1, 0)
99         c = math.ceil(p / 2)
100         return (0, 0), (0, 2), (1, c), (1, 2), (1, p - 1)
101     if q == 5:
102         if p == 3:
103             return (0, 0), (1, 1), (2, 2), (0, 1), (1, 2)
104         if p == 4:
105             return (0, 0), (0, 2), (2, 1), (0, 1), (0, 3)
106         if p == 5:
107             return (0, 0), (1, 1), (3, 3), (0, 2), (1, 0)
108         if p == 6:
109             return (0, 0), (1, 1), (3, 3), (0, 2), (1, 5)
110         if p == 7:
111             return (0, 0), (0, 4), (2, 1), (0, 3), (4, 1)
112         return (0, 0), (0, p // 2), (2, 0), (0, 2), (2, p - 1)
113     if q == 7:
114         if p == 3:
115             return (0, 0), (1, 1), (2, 2), (0, 1), (1, 2)
116         if p == 4:
117             return (0, 0), (0, 2), (2, 1), (0, 1), (0, 3)
118         return (0, 0), (0, 2), (3, 1), (1, 1), (5, 0)
119     r = (q - 1) // 2
120     if p == 3:

```

```

121         return (0, 0), (1, 1), (2, 2), (0, 1), (1, 2)
122     return (0, 0), (0, 2), (r, 0), (2, 0), (r + 3, 0)
123
124
125 def Kco_witness(p, q):
126     m = (p - 1) // 2
127     if p == 5 and q in (2, 3):
128         return (0, 0), (0, 1), (q - 1, 2), (0, 2), (1, 0)
129     if p == 5 and q >= 5:
130         return (0, 0), (1, 1), (4, 2), (0, 1), (3, 2)
131     if p >= 7 and 2 <= q <= p - 2:
132         j = max(2, m - q + 2)
133         return (0, 0), (0, 1), (q - 1, j), (0, m + 1), (1, 0)
134     if p == 7 and q >= 6:
135         return (0, 0), (0, 2), (0, 4), (0, 1), (0, 3)
136     if p >= 9 and q >= p - 1:
137         alpha = m - 3
138         beta = max(0, m - 5)
139         return (0, 0), (0, 2), (alpha, 4), (0, 1), (beta, 4)
140     raise ValueError("the exceptional case is Kco(5,4)")
141
142
143 def check_obstructions(P=25, Q=25):
144     for p in range(3, P + 1):
145         for q in range(3, Q + 1):
146             assert delta(Kh_graph(p, q), *Kh_witness(p, q)) == 1
147     for p in range(6, P + 1, 2):
148         for q in range(2, Q + 1):
149             w = (0, 0), (0, 1), (q, 0), (1, 0), (1, p // 2)
150             assert delta(Kce_graph(p, q), *w) == 1
151     for p in range(5, P + 1, 2):
152         for q in range(2, Q + 1):
153             if (p, q) != (5, 4):
154                 assert delta(Kco_graph(p, q), *Kco_witness(p, q)) == 1
155
156
157 def check_positive_example():
158     adj = Kco_graph(5, 4)
159     V = list(adj)
160     cuts = []
161     cuts.append({(0, j) for j in range(5)})
162     cuts.append({(3, j) for j in range(5)})
163     cuts.append({(i, j) for i in (0, 1) for j in range(5)})
164     for r in range(5):
165         cuts.append({(i, (r + t) % 5) for i in range(4) for t in range(4 - i)})
166     for k, x in enumerate(V):
167         D = bfs(adj, x)
168         for y in V[k + 1:]:
169             N = sum((x in C) ^ (y in C) for C in cuts)
170             assert N == 2 * D[y]
171
172
173 if __name__ == "__main__":
174     check_obstructions()
175     check_positive_example()
176     print("all listed obstruction certificates and the positive example checked")

```

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